

Aarati Kafle

Norman, Oklahoma

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Summary

Geospatial analyst and GIS researcher with strong experience in remote sensing, spatial data science, and environmental modeling. Skilled in developing reproducible GIS workflows for land cover mapping, agricultural monitoring, and land change studies using ArcGIS Pro, Google Earth Engine, Python, and R. Experienced in teaching GIS applications, collaborating on research projects, and presenting findings through high-quality maps, figures, and technical documentation.

Experience

Graduate Research Assistant

Jan 2025 – Present

Department of Geography & Environmental Sustainability, The University of Oklahoma

Norman, OK

- Conducting research on mapping tillage and non-tillage farming practices and evaluating water-use efficiency and crop productivity in winter wheat systems across Oklahoma using remote sensing and spatial analytics.
- Processing Landsat and Sentinel-2 imagery and deriving vegetation indices (NDVI/EVI) and crop-related metrics; integrating climate datasets (e.g., precipitation, temperature, ET) to support environmental interpretation.
- Developing geospatial workflows in ArcGIS Pro, Google Earth Engine, Python, and R for raster preprocessing, sampling, zonal statistics, and land cover classification.

Graduate Teaching Assistant – GIS Applications / Spatial Analysis

Aug 2024 – Dec 2025

The University of Oklahoma

Norman, OK

- Supported GIS lab instruction by guiding students through ArcGIS Pro workflows including geoprocessing, spatial joins, raster analysis, cartographic design, and spatial data management.
- Assisted with grading, lab material preparation, and troubleshooting student projects to strengthen understanding of GIS applications.

Geomatics / GIS Instructor

Jan 2023 – July 2024

Sharadpur Secondary School

Chitwan, Nepal

- Delivered training on GIS/remote sensing fundamentals including coordinate systems, map production, spatial analysis, and geospatial data interpretation.
- Supported 60+ students through hands-on projects using GIS tools and field-based mapping exercises.

GIS Analyst Intern

Nov 2021 – Jan 2022

Community Self Reliance Center

Kathmandu, Nepal

- Conducted spatial analysis for land rights and community mapping.
- Processed and analyzed geospatial datasets using ArcGIS and QGIS.

Skills

GIS/Remote Sensing: ArcGIS Pro, ArcGIS Online, QGIS, Google Earth Engine (GEE), ModelBuilder, ENVI,
Programming: Python (ArcPy, GeoPandas, Rasterio, NumPy, Pandas), R (sf, terra/raster, ggplot2), JAVA, C, C++
Spatial Methods: Land cover classification (RF), change detection, raster processing, hotspot analysis, map design, and analysis
Interpersonal: Teamwork, Leadership, Public Speaking, Conflict Resolution, Collaboration
Other: Cartography, Data QA/QC, Scientific writing

Education

Master of Science in Geography & Environmental Sustainability

Aug 2024 – May 2026 (Expected)

The University of Oklahoma

Norman, OK

- Thesis: *Assessing the Impact of Tillage and Non-Tillage Farming Practices and Analyzing Water-Use Efficiency and Crop Productivity in Winter Wheat Fields in Oklahoma*

Bachelor's Degree in Geomatics Engineering

Aug 2017 – June 2022

Kathmandu University

Dhulikhel, Nepal

- Thesis: *Web Map Application for Preliminary Flood Damage Assessment in Melamchi and Indrawati Municipality, Nepal.*
- Relevant coursework: GIS, Remote Sensing, Photogrammetry, Surveying, Spatial Analysis, Cartography

Projects

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- Assessing Agricultural Land Cover Change in Oklahoma Using Cropland Data Layer (CDL)** 2025
- Analyzed multi-year USDA CDL raster datasets (2015–2024) to quantify agricultural land cover patterns and temporal change across Oklahoma.
 - Reclassified detailed CDL categories into simplified classes (e.g., winter wheat, pasture/grassland, developed, forest, water) and computed class-wise area statistics.
 - Developed maps and visual summaries to identify agricultural conversion trends, dominant crops, and temporal transitions.
- Assessment of Urbanization Patterns and Land Cover Changes in Oklahoma** 2024
- Evaluated urban expansion and land cover transitions using GIS/remote sensing change detection workflows.
 - Created land cover transition maps and spatial summaries to highlight urban growth hotspots.
- Analysis of Evapotranspiration (ET) and Land Surface Temperature (LST) in Oklahoma, USA** 2024
- Examined ET–LST relationships using spatial datasets, raster processing, and geostatistical workflows.
 - Produced thematic maps and statistical summaries to interpret spatial variability and climate–land surface interactions.
- UAV-Derived DEM vs. Total Station Survey DEM (Dhulikhel, Kavre, Nepal)** 2022
- Compared UAV-derived DEM outputs with total station survey DEM to evaluate terrain representation and elevation accuracy.
 - Calculated error metrics (RMSE) and visualized spatial error distribution across complex topography.
- Suitability Analysis for Waste Disposal Site (Chitwan District, Nepal)** 2021
- Applied GIS-based multi-criteria decision analysis integrating slope, land use, proximity to roads/settlements/water bodies, and protected areas.
 - Produced suitability maps (high/moderate/low) to support evidence-based planning and decision-making.
- Burn Area Assessment of Myagdi District (Nepal) Using Remote Sensing** 2021
- Mapped burn extent and severity using satellite imagery and indices such as NDVI, NBR, and dNBR for environmental monitoring.
 - Estimated burned area coverage and generated thematic maps for post-fire assessment and restoration planning.

Awards & Scholarships

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- Second Place — GIS Day Graduate Student Poster Competition, The University of Oklahoma (2025)
 - Pugh Family Pioneer Scholarship, The University of Oklahoma (2025)
 - College of A&GS Research Travel Assistance Award, The University of Oklahoma (2025)
 - DGES Travel Award, The University of Oklahoma (2025)
 - GeoBowl SWAAG — Runner-Up (2024)
 - Facebook Community Impact Microgrant, Humanitarian OpenStreetMap (2021)

Poster Presentations

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- **Kafle, A.**, Bhattarai, N., Wagle, P., Paudel, J., Sharma, S., & Woznicki, S. A. (2025). Geospatial mapping of tillage and no-tillage farming practices and their impacts on winter wheat water-use efficiency in Oklahoma. Poster presented at GIS Day, University of Oklahoma, Norman, OK.
 - **Kafle, A.**, Bhattarai, N., Wagle, P., Paudel, J., Sharma, S., & Woznicki, S. A. (2025). Mapping No tillage farming impacts on winter wheat water-use efficiency in Oklahoma. Poster presented at Governor’s Water Conference Research Symposium, Norman, OK.
 - **Kafle, A.**, Bhattarai, N., Wagle, P., Paudel, J., Sharma, S., & Woznicki, S. A. (2025). Assessing the impact of tillage and no-tillage farming practices on winter wheat productivity and water-use efficiency in Oklahoma. Poster presented at the American Geophysical Union (AGU) Fall Meeting, New Orleans, LA.

Volunteering & Leadership

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- Volunteer, OU Booth — American Geophysical Union (AGU) Annual Meeting (2025).
 - GIS Day Volunteer, The University of Oklahoma (2024)
 - Community Mapping Volunteer, Humanitarian OpenStreetMap (2021–2022)
 - Organizer, Nepal Geomatics Meet (NepGeom) 2019 — Kathmandu University
 - Executive Member, Geomatics Engineering Society 2018 — Kathmandu University